Condition-Aware Cross-Modal Adaptive Fusion for Drone-View Multimodal Semantic Segmentation

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Abstract

Drone-view multimodal semantic segmentation faces two coupled challenges. First, the relative reliability of RGB and thermal modalities varies substantially across illumination, viewpoint, and weather conditions, making static fusion strategies brittle. Second, the bird's-eye viewpoint severely compresses object scale, so safety-critical categories such as pedestrians, traffic lights, and motorcycles are easily suppressed during downsampling and decoding. To address these issues, we propose CACAF-HGSOAD, a foundation-model-driven RGB-T segmentation framework for drone scenes. The framework introduces Condition-Aware Cross-Modal Adaptive Fusion (CACAF), which estimates modality quality and performs adaptive cross-modal enhancement, and a Hierarchical Guided Small-Object-Aware Decoder (HGSOAD), which combines hierarchical decoding, scale-aware gating, and auxiliary supervision to preserve fine-grained structures. Built on a frozen SAM2 Hiera-L RGB backbone with lightweight bottleneck adapters and a fully trainable ConvNeXt-Tiny thermal encoder, the proposed design balances strong pre-trained semantics and task-specific multimodal adaptation.

On the FMB test set, CACAF-HGSOAD achieves **68.62 mIoU**, outperforming MM SAM-Adapter by **+2.52**. Under a unified training protocol, removing CACAF causes a **5.76 mIoU** drop and removing SAGU causes a **3.05 mIoU** drop, validating the contribution of both proposed components. Loss ablations further show that **CE + Dice** is consistently stronger than **CE + OHEM** and **CE + Dice + OHEM**. Under the strict 9-class MFNet protocol, the proposed method reaches **54.73 mIoU**, outperforming MFNet and RTFNet while remaining competitive with stronger recent baselines.

1 Introduction

Multimodal semantic segmentation improves scene understanding by combining complementary sensing cues such as RGB, depth, thermal, and other auxiliary modalities. This direction has become increasingly important for robust perception under adverse illumination, occlusion, and environmental uncertainty [1, 2, 3]. For drone platforms, however, multimodal segmentation is particularly challenging. First, the bird's-eye viewpoint dramatically reduces object scale, making small categories easy to lose during repeated downsampling and decoding. Second, changes in altitude, orientation, and lighting conditions make the relative usefulness of RGB and thermal modalities highly dynamic. As a result, fixed fusion strategies often fail to exploit cross-modal complementarity in a scene-dependent manner. Existing UAV-oriented datasets and models, such as UAVid, Agriculture-Vision, SynDrone, UAVformer, and 2DSegFormer, also indicate that aerial scene parsing differs substantially from ground-view segmentation in scale distribution, long-range context, and structural recovery [4, 5, 6, 7, 8].

Recent foundation models have substantially strengthened visual representation learning. Large-scale pre-training with ViT, MAE, and ConvNeXt has established powerful generic backbones [9, 10, 11]. Building on this trend, SAM and SAM 2 provide strong generic segmentation priors, while ViT-Adapter, SAM-Adapter, SAM2-Adapter, and MM SAM-Adapter demonstrate that frozen foundation backbones with lightweight adaptation modules form an effective route to dense prediction and multimodal transfer [12, 13, 14, 15, 16, 17]. At the same time, LoRA-style parameter-efficient tuning strategies and their vision variants further support the design choice of keeping most pre-trained parameters frozen while adapting only a small trainable subset [18, 19, 20]. Nevertheless, for drone-view RGB-T segmentation on FMB, two limitations remain clear. Existing methods usually do not explicitly model modality-quality variation at the input-feature level, and their decoders are rarely designed around the severe small-object degradation caused by aerial viewpoints.

To address these limitations, we propose CACAF-HGSOAD, a foundation-model-driven RGB-T segmentation framework tailored to drone scenes. The framework uses a frozen SAM2 Hiera-L RGB backbone with bottleneck adapters and a fully trainable ConvNeXt-Tiny thermal encoder in an asymmetric dual-stream design. On top of these encoders, we introduce Condition-Aware Cross-Modal Adaptive Fusion (CACAF), which estimates modality quality and performs adaptive cross-modal enhancement at multiple scales. We further propose the Hierarchical Guided Small-Object-Aware Decoder (HGSOAD), which combines hierarchical decoding, scale-aware gating, and auxiliary supervision to improve the recovery of fine-grained structures and distant small objects. The overall design is centered on FMB, a full-time RGB-T benchmark that is especially suitable for evaluating robustness to modality-quality fluctuations in challenging aerial conditions [21, 17].

Our main contributions are summarized as follows:

1. We propose CACAF, a condition-aware fusion module that explicitly estimates modality quality and performs adaptive cross-modal enhancement instead of relying on static multimodal fusion.
2. We propose HGSOAD with SAGU, a lightweight decoder enhancement tailored to the small-object recovery problem in drone-view segmentation.
3. On the FMB test set, the proposed method achieves **68.62 mIoU**, improving over MM SAM-Adapter by **+2.52**.
4. We provide unified module and loss-function ablations, clarifying the individual roles of CACAF, SAGU, and the CE + Dice training objective.

2 Related Work

2.1 Multimodal Semantic Segmentation

RGB-X semantic segmentation improves robustness by incorporating thermal, depth, polarization, LiDAR, or other auxiliary modalities, and comprehensive overviews can be found in prior surveys [1]. Early RGB-T methods mainly relied on dual-branch CNN encoders and explicit fusion decoders, as exemplified by MFNet, RTFNet, and FuseNet [22, 23, 2]. Later work explored stronger feature interaction through self-supervised adaptation, non-local fusion, attention-based fusion, complementarity-aware modeling, and real-time multimodal decoding [24, 25, 26, 27, 28]. More recent methods, including GMNet, CMX, CMNeXt, Multimodal Token Fusion, GeminiFusion, RoadFormer, RoadFormer+, and CAFuser, move further toward transformer-based or condition-aware cross-modal interaction [29, 3, 30, 31, 32, 33, 34, 35]. Despite this progress, most of these methods focus on driving or generic road scenes rather than drone-view perception with large scale variation and severe modality-quality fluctuations.

2.2 Foundation Model Adaptation for Dense Prediction

The success of large-scale visual pre-training has made foundation-model adaptation a central topic in dense prediction. ViT and MAE established strong transformer-based visual backbones, while ConvNeXt showed that modernized ConvNets can remain highly competitive [9, 10, 11]. ViT-Adapter demonstrated that dense prediction performance can be substantially improved by attaching lightweight trainable modules to frozen pre-trained transformers [14]. In parallel, parameter-efficient tuning methods such as LoRA, Conv-LoRA, and recent PEFT surveys highlight the practical value of updating only a small set of trainable parameters when adapting large pre-trained models [18, 19, 20]. For segmentation foundation models, SAM-Adapter and SAM2-Adapter extended this philosophy to SAM and SAM 2, while MM SAM-Adapter further injected auxiliary-modal information into a SAM-based RGB backbone for multimodal semantic segmentation [15, 16, 17]. Our work follows this adaptation route, but targets drone-view RGB-T segmentation and emphasizes condition-aware fusion and small-object-aware decoding.

2.3 Drone-View Scene Parsing

Compared with ground-view RGB-T segmentation, drone-view segmentation involves stronger perspective distortion, broader scale variation, and much more severe small-object degradation. UAVid, Agriculture-Vision, and SynDrone illustrate these characteristics from different aerial or UAV scenarios [4, 5, 6]. Architectures such as UAVformer and 2DSegFormer explicitly target aerial semantic segmentation with stronger global modeling [7, 8]. On the remote-sensing side, RS-SAM, Multiscale Adapter, and SAM2Former investigate how SAM-family models can be adapted to high-resolution aerial and remote-sensing segmentation [36, 37, 38]. However, these methods mainly focus on RGB imagery or remote-sensing semantics rather than RGB-T fusion under drone-specific modality instability. Our design therefore focuses on two aspects that remain under-addressed in this setting: condition-dependent multimodal fusion and the recovery of small targets in hierarchical decoding.

3 Method

3.1 Problem Formulation and Overview

Given an RGB image I^{rgb}, a thermal image I^{thm}, and a pixel-wise annotation map Y, the goal is to learn a mapping

$$\mathcal{F} : (I^{rgb}, I^{thm}) \mapsto \hat{Y}, \tag{1}$$

where $\hat{Y}$ denotes the semantic segmentation prediction. Our design targets two coupled difficulties in drone-view RGB-T segmentation: dynamic modality-quality variation and severe loss of small objects.

As shown in Fig. 1, the framework consists of four components: a frozen RGB backbone, a trainable thermal encoder, multi-scale CACAF fusion blocks, and the HGSOAD decoder. The RGB branch uses SAM2 Hiera-L as the main semantic prior source [13], while the thermal branch uses ConvNeXt-Tiny to maintain adaptability to low-light and degraded conditions [11]. Features from both branches are fused at four scales through CACAF and then decoded in a top-down manner by HGSOAD.

3.2 RGB and Thermal Encoders

We use SAM2 Hiera-L as the RGB backbone and freeze its original parameters, training only bottleneck adapters inserted before the Hiera blocks. This follows the adaptation strategy of ViT-Adapter and SAM-family adapters, where a strong pre-trained backbone is preserved and only a small trainable branch is introduced for downstream transfer [14, 15, 16]. For an

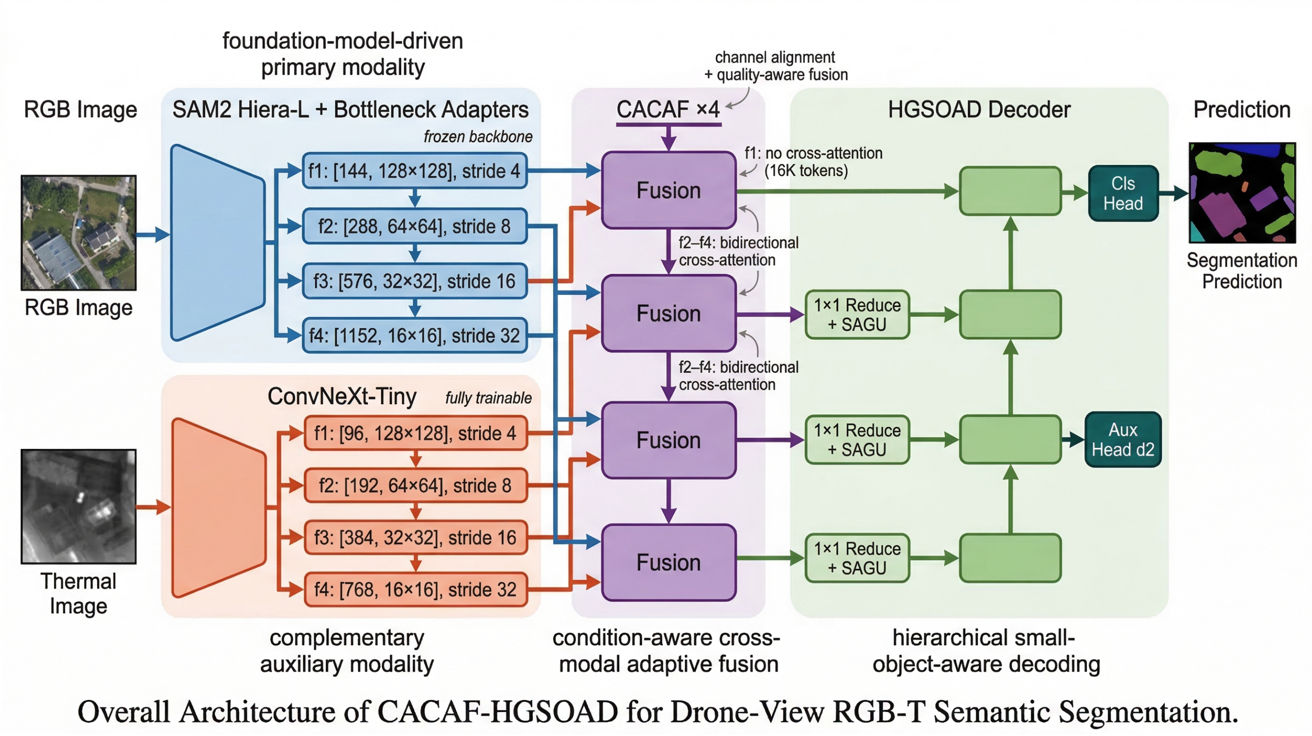

Figure 1: Overall architecture of CACAF-HGSOAD. The framework adopts an asymmetric dual-encoder design, performs condition-aware cross-modal fusion at four scales, and decodes the fused features with a hierarchical small-object-aware decoder.

input token x, the adapter output is

$$\mathrm{Adapter}(x) = W_2(\mathrm{GELU}(W_1 x)). \tag{2}$$

The transformed token fed into the original block becomes $x + \mathrm{Adapter}(x)$.

The thermal branch uses a fully trainable ConvNeXt-Tiny encoder [11] to extract four-level multi-scale features. This yields an asymmetric multimodal design: the RGB stream provides strong foundation-model semantics, while the thermal stream remains flexible enough to absorb dataset- and condition-specific information.

3.3 CACAF

For RGB features f_i^{rgb} and thermal features f_i^{thm} at the i-th scale, CACAF performs four steps.

1. **Channel alignment**

$$\hat{f}_i^{rgb} = \psi_i^{rgb}(f_i^{rgb}), \quad \hat{f}_i^{thm} = \psi_i^{thm}(f_i^{thm}). \tag{3}$$

2. **Modal quality assessment**

$$q_i^{rgb} = \mathrm{MLP}(\mathrm{GAP}(\hat{f}_i^{rgb})), \tag{4}$$
$$q_i^{thm} = \mathrm{MLP}(\mathrm{GAP}(\hat{f}_i^{thm})). \tag{5}$$

followed by

$$[w_i^{rgb}, w_i^{thm}] = \mathrm{softmax}([q_i^{rgb}, q_i^{thm}]). \tag{6}$$

3. **Cross-modal enhancement** For the highest-resolution $f1$ level, we retain quality-aware weighting but omit cross-attention for efficiency. For $f2/f3/f4$, we apply bidirectional multi-head attention to exchange cross-modal information.

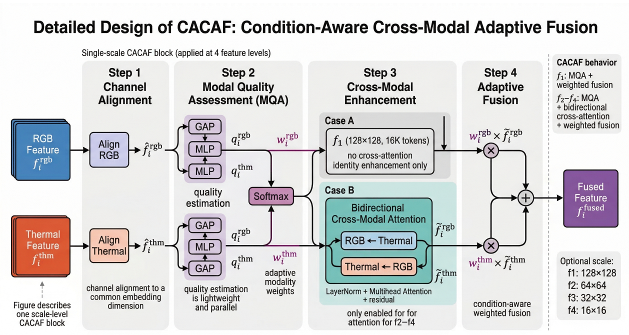

Figure 2: Detailed design of CACAF. At each scale, modality features are aligned to a shared embedding space, weighted by modal quality assessment, enhanced by bidirectional cross-modal attention when affordable, and finally fused with adaptive weights.

4. **Adaptive fusion**

$$f_i^{fused} = w_i^{rgb} \tilde{f}_i^{rgb} + w_i^{thm} \tilde{f}_i^{thm}. \tag{7}$$

Fig. 2 illustrates the internal flow of CACAF. Rather than simply concatenating or averaging modality features, CACAF first aligns them in a shared embedding space, then predicts quality-aware modality weights, and finally performs explicit cross-modal interaction when the token count is computationally manageable. This allows the model to respond to scene-dependent changes in modality reliability.

3.4 HGSOAD

HGSOAD adopts a U-Net-style top-down decoder. The fused features at four scales are first projected to a common decoding dimension through 1×1 convolutions, then progressively upsampled and merged through skip connections. At each scale, HGSOAD applies a Scale-Aware Gating Unit (SAGU) for lightweight channel gating:

$$g_i = \sigma\left(\phi(\mathrm{GAP}(x_i)) + \phi(\mathrm{GMP}(x_i))\right), \tag{8}$$

$$\hat{x}_i = g_i \odot x_i, \tag{9}$$

where ϕ denotes a 1×1 convolution, followed by ReLU and another 1×1 convolution. This gating mechanism improves the decoder's ability to preserve channels that are useful for fine structures and distant small targets.

3.5 Training Objective

We use a CE + Dice objective on the main head and auxiliary heads:

$$\mathcal{L} = \mathcal{L}_{main} + \lambda\left(\mathcal{L}_{aux2} + \mathcal{L}_{aux3}\right), \tag{10}$$

where

$$\mathcal{L}_{main} = \mathcal{L}_{CE}(p, y) + \mathcal{L}_{Dice}(p, y), \tag{11}$$

$$\mathcal{L}_{auxk} = \mathcal{L}_{CE}(p_k^{aux}, y) + \mathcal{L}_{Dice}(p_k^{aux}, y), \quad k \in \{2, 3\}, \tag{12}$$

and $\lambda = 0.4$. As shown later in the experiments, combining Dice and OHEM is not beneficial in this architecture, so the final model is trained with CE + Dice.

4 Experiments

4.1 Dataset and Setup

FMB is a real-world RGB-T semantic segmentation benchmark with 14 semantic classes and a train/val/test split of 1060/160/280 [21]. The original test resolution is 800×600. Compared with aerial RGB segmentation datasets such as UAVid and Agriculture-Vision, FMB additionally emphasizes all-day RGB-T paired observations, making it more suitable for evaluating modality-quality variation in multimodal aerial perception [4, 5, 21]. During training, we use random scaling and random cropping to generate 512×512 inputs. For the main test protocol, inference is performed at the original image resolution, padding inputs only to multiples of 32 instead of forcing global resizing to 512×512.

We use AdamW for optimization. The learning rates for the SAM2 adapters and ConvNeXt-Tiny are both set to 1×10^{-4}, while CACAF and HGSOAD use 2×10^{-4}. The weight decay is 0.01. We adopt a schedule with 500 warmup iterations followed by cosine decay. The model is trained for 60 epochs with BF16 mixed precision and batch size 4. Data augmentation includes random scaling, random cropping, random horizontal flipping, Gaussian blur ($p = 0.2$), photometric distortion, and a category-aware crop constraint with $cat\_max\_ratio = 0.75$.

Table 1: Comparison with representative methods on the FMB test set. Results are taken from official papers or official repositories released by the cited methods.

Method	mIoU
MMSFormer [39]	61.68
StitchFusion [40]	64.85
MM SAM-Adapter [17]	66.10
Ours (CACAF-HGSOAD)	**68.62**

Table 2: Module ablation on the FMB test set.

Method	mIoU	Δ
Full (CACAF + SAGU)	**68.62**	—
w/o CACAF	62.86	-5.76
w/o SAGU	65.57	-3.05
w/o CACAF, w/o SAGU	64.78	-3.84

4.2 Comparison with Prior Methods

Table 1 compares the proposed method with representative published or officially released FMB baselines. Our method surpasses MMSFormer [39], StitchFusion [40], and the direct baseline MM SAM-Adapter [17], achieving the best mIoU among the currently verified entries. Relative to MM SAM-Adapter, the improvement is **+2.52 mIoU**; in our local evaluation, the corresponding mAcc gain is **+3.66**, while aAcc decreases slightly from 93.95 to 93.27.

4.3 Module and Loss Ablations

The ablation results show that CACAF is the dominant source of improvement, while SAGU provides an additional and consistent gain on top of the full architecture. This confirms that both adaptive multimodal fusion and decoder-side small-object enhancement are necessary for the drone-view RGB-T setting. The loss ablation further shows that Dice and OHEM do not complement each other in the current architecture; instead, their combination degrades performance. Consequently, the final model uses CE + Dice.

4.4 Cross-Dataset Evaluation

Under the strict 9-class MFNet protocol, the proposed method achieves **54.73 mIoU**, **68.14 mAcc**, and **97.90 aAcc**. It clearly outperforms MFNet [22] and RTFNet [23], remains close to EGFNet [41], but trails stronger transformer-based methods such as CMX [3], CMNeXt [30], StitchFusion [40], and CRM [42]. These results suggest that the proposed design is not limited to FMB, while also indicating that there remains room for stronger cross-dataset generalization.

4.5 Qualitative Analysis

Fig. 3 presents five representative FMB test cases, each with RGB, thermal, ground-truth, MM SAM-Adapter, and our prediction. The proposed model more reliably

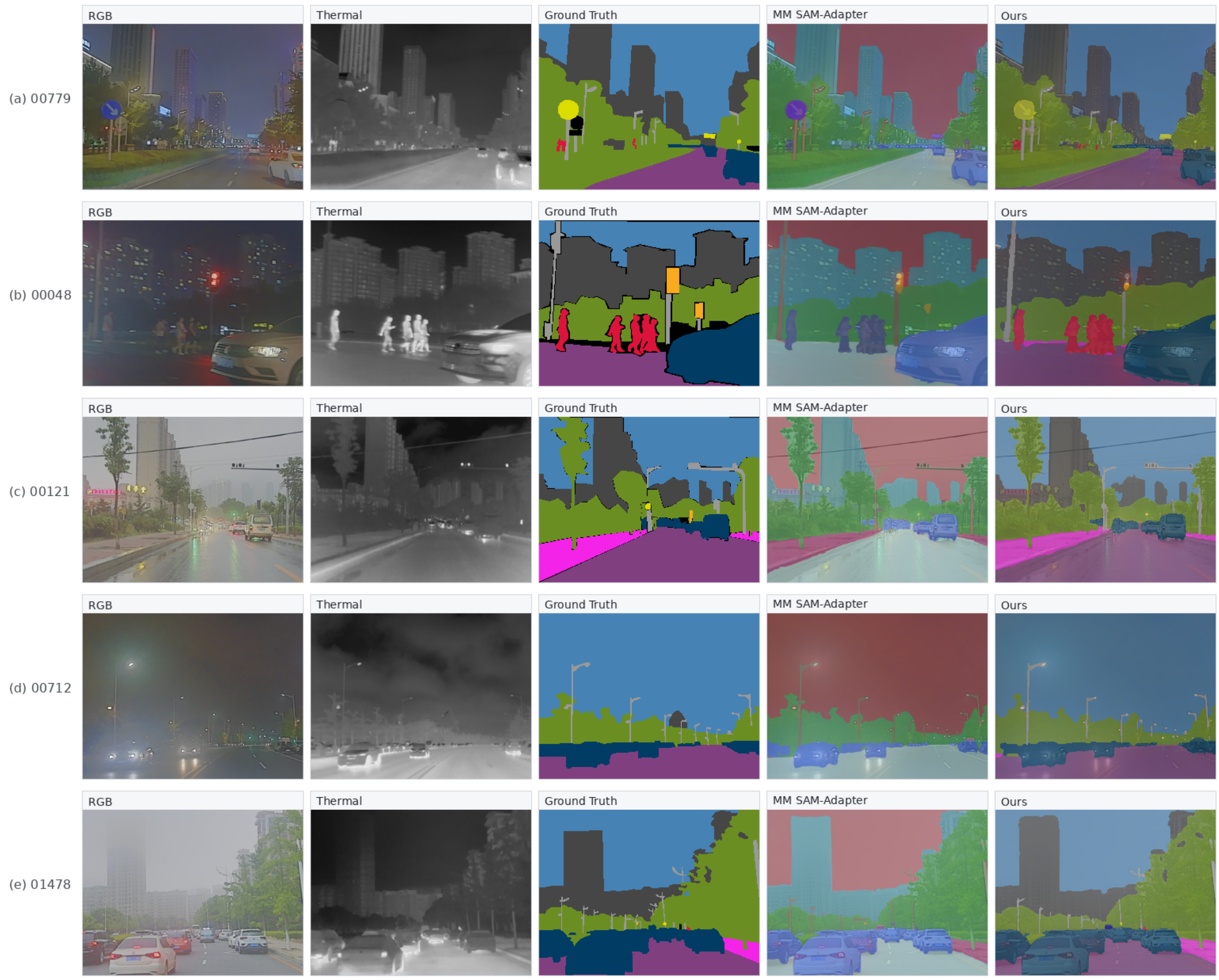

Figure 3: Qualitative comparisons on five representative FMB test scenes (`00779`, `00048`, `00121`, `00712`, and `01478`). Compared with MM SAM-Adapter, the proposed method yields clearer boundaries and more reliable recovery of small or weakly illuminated targets under challenging RGB-T conditions.

Table 3: Loss ablation on the FMB test set.

Loss	mIoU	Δ
CE + Dice	**68.62**	—
CE + OHEM	67.79	-0.83
CE + Dice + OHEM	66.79	-1.83

recovers pedestrians, traffic lights, and distant vehicles in low-light or low-contrast scenes, while also producing cleaner road boundaries and object contours. Fig. 4 provides an interpretability view of CACAF. The learned fusion weights show that thermal cues contribute more clearly at shallow and intermediate scales, where local appearance degradation and low-contrast regions are still informative for cross-modal compensation. In contrast, deeper scales are largely dominated by the RGB branch, indicating that high-level semantic representations rely more heavily on the frozen SAM2 backbone. This observation suggests that the main benefit of CACAF lies in condition-aware adaptation at early and middle stages, rather than uniformly strong dynamic weighting at all scales.

Table 4: MFNet comparison under the strict 9-class protocol.

Method	mIoU
MFNet [22]	45.1
RTFNet [23]	53.2
EGFNet [41]	54.8
GMNet [29]	57.3
CMX (MiT-B4) [3]	59.7
CMNeXt (MiT-B4) [30]	59.9
StitchFusion [40]	58.13
CRM (Swin-B) [42]	**61.4**
Ours (CACAF-HGSOAD)	54.73

5 Conclusion

We presented CACAF-HGSOAD for drone-view RGB-T semantic segmentation. The proposed framework preserves the strong "frozen foundation backbone + lightweight adaptation" paradigm while introducing condition-aware cross-modal fusion and a small-object-oriented hierarchical decoder. On the FMB test set, it achieves **68.62 mIoU**, improving over MM SAM-Adapter by **+2.52**. Unified ablations confirm the contributions of CACAF and SAGU, while loss ablations identify CE + Dice as the most suitable objective for the current architecture. Supplemental MFNet results further indicate that the design retains nontrivial transferability beyond the primary benchmark.

References

[1] Y. Zhang, D. Sidibé, O. Morel, and F. Mériaudeau, "Deep multimodal fusion for semantic image segmentation: A survey," *Image and Vision Computing*, vol. 105, p. 104042, 2021.

[2] C. Hazirbas, L. Ma, C. Domokos, and D. Cremers, "FuseNet: Incorporating depth into semantic segmentation via fusion-based CNN architecture," in *Proceedings of the Asian Conference on Computer Vision*, pp. 213–228, 2016.

[3] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 12, pp. 14679–14694, 2023.

[4] Y. Lyu, X. Bai, C. Liao, *et al.*, "UAVid: A semantic segmentation dataset for UAV imagery," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 165, pp. 108–119, 2020.

[5] M.-J. Chiu, X. Xu, Y. Wang, *et al.*, "Agriculture-vision: A large aerial image database for agricultural pattern analysis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2828–2838, 2020.

[6] G. Rizzoli, F. Poiesi, E. Ricci, *et al.*, "SynDrone: Multi-modal UAV dataset for urban scenarios," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2023.

[7] Y. Zhang, Z. Chen, H. Hu, *et al.*, "UAVformer: A composite transformer network for urban scene segmentation of UAV images," *Pattern Recognition*, vol. 138, p. 109406, 2023.

[8] R. Li, J. Cao, Q. Fu, J. Li, J. Sun, and T. Tian, "2DSegFormer: 2-d transformer model for semantic segmentation on aerial images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–13, 2022.

[9] A. Dosovitskiy, L. Beyer, A. Kolesnikov, *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proceedings of the International Conference on Learning Representations*, 2021.

[10] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15979–15988, 2022.

[11] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11976–11986, 2022.

[12] A. Kirillov, E. Mintun, N. Ravi, *et al.*, "Segment anything," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4015–4026, 2023.

[13] N. Ravi, V. Gabeur, Y.-T. Hu, *et al.*, "SAM 2: Segment anything in images and videos," in *Proceedings of the International Conference on Learning Representations*, 2025.

[14] Z. Chen, Y. Duan, W. Wang, *et al.*, "Vision transformer adapter for dense predictions," in *Proceedings of the International Conference on Learning Representations*, 2023.

[15] T. Chen, L. Zhu, C. Ding, *et al.*, "SAM-adapter: Adapting segment anything in underperformed scenes," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pp. 3367–3375, 2023.

[16] T. Chen, A. Lu, L. Zhu, *et al.*, "SAM2-adapter: Evaluating and adapting segment anything 2 in downstream tasks," 2024.

[17] I. Curti, P. Z. Ramirez, A. Petrelli, and L. D. Stefano, "Multimodal SAM-adapter for semantic segmentation," *IEEE Access*, vol. 13, pp. 160438–160455, 2025.

[18] E. J. Hu, Y. Shen, P. Wallis, *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proceedings of the International Conference on Learning Representations*, 2022.

[19] Z. Zhong, M. Cheng, X. Chen, *et al.*, "Convolution meets LoRA: Parameter efficient finetuning for segment anything model," in *Proceedings of the International Conference on Learning Representations*, 2024.

[20] Y. Xin, J. Yang, S. Luo, *et al.*, "Parameter-efficient finetuning for pre-trained vision models: A survey and benchmark," 2024.

[21] J. Liu, Z. Liu, G. Wu, L. Ma, R. Liu, W. Zhong, Z. Luo, and X. Fan, "Multi-interactive feature learning and a full-time multi-modality benchmark for image fusion and segmentation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 8081–8090, 2023.

[22] Q. Ha, K. Watanabe, T. Karasawa, Y. Ushiku, and T. Harada, "MFNet: Towards real-time semantic segmentation for autonomous vehicles with multi-spectral scenes," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, pp. 5108–5115, 2017.

[23] Y. Sun, W. Zuo, and M. Liu, "RTFNet: RGB-thermal fusion network for semantic segmentation of urban scenes," *IEEE Robotics and Automation Letters*, vol. 4, no. 3, pp. 2576–2583, 2019.

[24] A. Valada, R. Mohan, and W. Burgard, "Self-supervised model adaptation for multimodal semantic segmentation," *International Journal of Computer Vision*, vol. 128, no. 5, pp. 1239–1285, 2020.

[25] X. Yan, K. Wang, Y. Huang, *et al.*, "NLFNet: Non-local fusion towards generalized multimodal semantic segmentation across RGB-depth, polarization, and thermal images," in *Proceedings of the IEEE International Conference on Robotics and Biomimetics*, pp. 1748–1753, 2021.

[26] D. Xu, H. Zhao, X. Kong, *et al.*, "Attention fusion network for multi-spectral semantic segmentation," *Pattern Recognition Letters*, vol. 146, pp. 74–80, 2021.

[27] Z. Wu, Y. Liu, L. Wang, *et al.*, "Complementarity-aware cross-modal feature fusion network for RGB-t semantic segmentation," *Pattern Recognition*, vol. 128, p. 108693, 2022.

[28] Y. Sun, W. Zuo, and M. Liu, "Real-time fusion network for RGB-d semantic segmentation incorporating unexpected obstacle detection for road-driving images," *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 5558–5565, 2020.

[29] W. Zhou, S. Liang, J. Yang, *et al.*, "GMNet: Graded-feature multilabel-learning network for RGB-thermal urban scene semantic segmentation," *IEEE Transactions on Image Processing*, vol. 30, pp. 7790–7802, 2021.

[30] J. Zhang, R. Liu, H. Shi, *et al.*, "Delivering arbitrary-modal semantic segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 1136–1147, 2023.

[31] Y. Wang, X. Chen, L. Cao, W. Huang, F. Sun, and Y. Wang, "Multimodal token fusion for vision transformers," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12176–12185, 2022.

[32] D. Jia, J. Guo, K. Han, *et al.*, "GeminiFusion: Efficient pixel-wise multimodal fusion for vision transformer," in *Proceedings of the International Conference on Machine Learning*, pp. 21753–21767, 2024.

[33] J. Li, Y. Zhang, P. Yun, G. Zhou, Q. Chen, and R. Fan, "RoadFormer: Duplex transformer for RGB-normal semantic road scene parsing," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 7, pp. 5163–5172, 2024.

[34] J. Huang, J. Li, N. Jia, *et al.*, "RoadFormer+: Delivering RGB-X scene parsing through scale-aware information decoupling and advanced heterogeneous feature fusion," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 5, pp. 3156–3165, 2025.

[35] T. Broedermann, C. Sakaridis, Y. Fu, and L. V. Gool, "CAFuser: Condition-aware multimodal fusion for robust semantic perception of driving scenes," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3134–3141, 2025.

[36] E. Zhang, J. Liu, A. Cao, Z. Sun, H. Zhang, H. Wang, L. Sun, and M. Song, "RS-SAM: Integrating multi-scale information for enhanced remote sensing image segmentation," in *Proceedings of the Asian Conference on Computer Vision*, 2024.

[37] S. Chen, Y. Yu, Y. Li, Z. Wang, X. Li, and J. Han, "Multiscale adapter based on SAM for remote sensing semantic segmentation," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 18, pp. 6806–6824, 2025.

[38] X. Li, X. Tian, Z. Wang, F. Zhang, Y. Zhang, N. Yang, and C. Tian, "SAM2Former: Segment anything model 2 assisting UNet-like transformer for remote sensing image semantic segmentation," *IEEE Access*, vol. 13, pp. 119212–119229, 2025.

[39] M. K. Reza, A. Prater-Bennette, and M. S. Asif, "MMSFormer: Multimodal transformer for material and semantic segmentation," *IEEE Open Journal of Signal Processing*, vol. 5, pp. 599–610, 2024.

[40] B. Li, D. Zhang, Z. Zhao, J. Gao, and X. Li, "StitchFusion: Weaving any visual modalities to enhance multimodal semantic segmentation," 2024.

[41] W. Zhou, S. Dong, C. Xu, and Y. Qian, "EGFNet: Edge-aware guidance fusion network for RGB-thermal scene parsing," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, pp. 3571–3579, 2022.

[42] U. Shin, K. Lee, I. S. Kweon, and J. Oh, "Complementary random masking for RGB-thermal semantic segmentation," 2023.

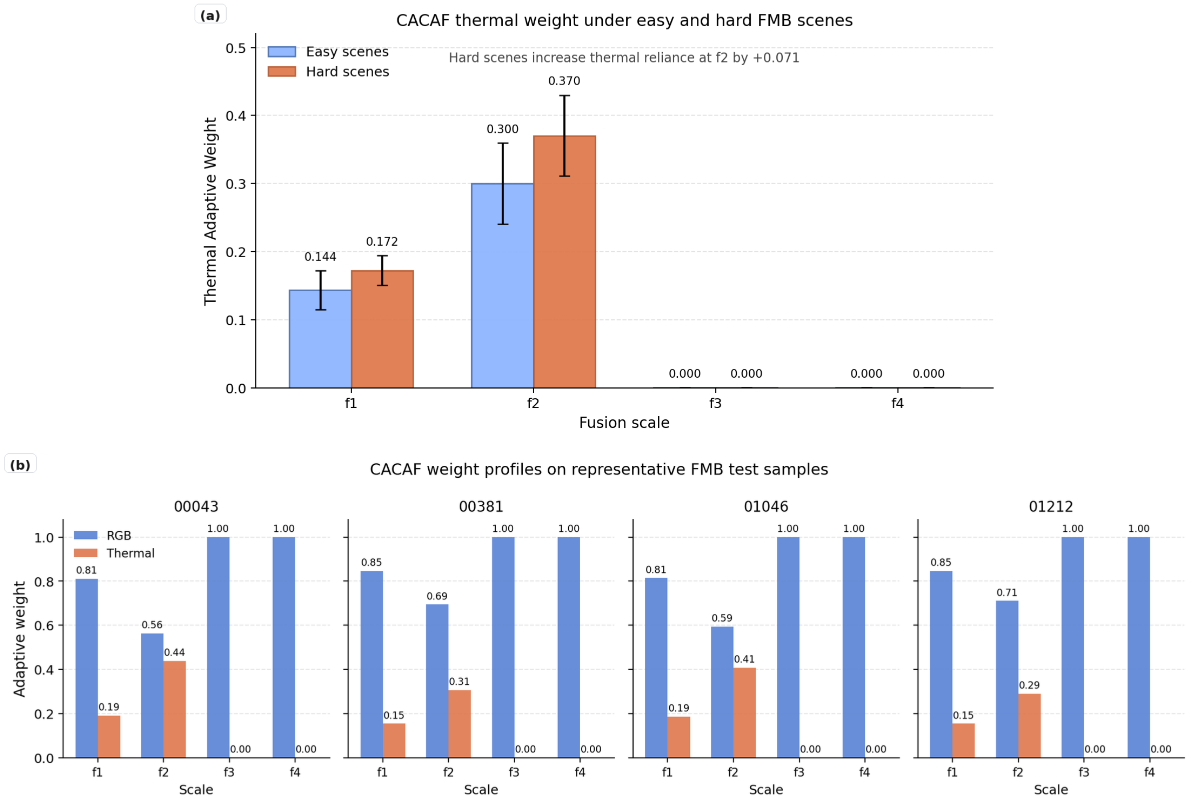

Figure 4: Visualization of adaptive fusion weights in CACAF. Thermal contributions are more evident at shallow and intermediate scales, especially in challenging conditions, while deeper scales are largely dominated by the RGB branch. This suggests that condition-aware multimodal adaptation mainly occurs before high-level semantic representations become strongly RGB-centered.